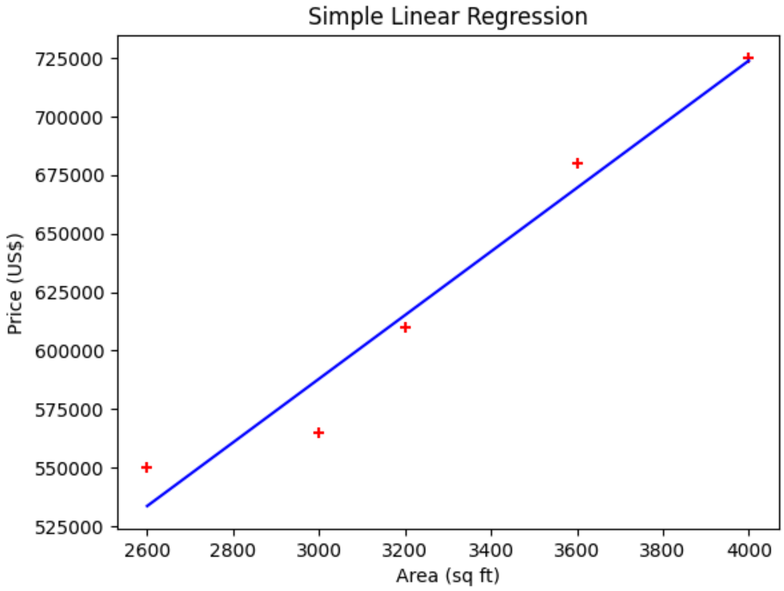
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

data = {
    'Area': [2600, 3000, 3200, 3600, 4000],
    'Price': [550000, 565000, 610000, 680000, 725000]
}
df1 = pd.DataFrame(data)

print("--- TASK 1: SIMPLE LINEAR REGRESSION ---")
print(df1)

X = df1[['Area']]
y = df1['Price']

reg = LinearRegression()
reg.fit(X, y)

print(f"\nCoefficient (Slope m): {reg.coef_[0]:.2f}")
print(f"Intercept (c): {reg.intercept_:.2f}")
test_area = [[3300]]
predicted_price = reg.predict(pd.DataFrame([[3300]], columns=['Area']))
print(f"Prediction for 3300 sq ft: ${predicted_price[0]:.2f}")
plt.xlabel('Area (sq ft)')
plt.ylabel('Price (US$)')
plt.scatter(df1.Area, df1.Price, color='red', marker='+')
plt.plot(df1.Area, reg.predict(X), color='blue')
plt.title("Simple Linear Regression")
plt.show()
```

```
--- TASK 1: SIMPLE LINEAR REGRESSION ---
   Area   Price
0  2600  550000
1  3000  565000
2  3200  610000
3  3600  680000
4  4000  725000

Coefficient (Slope m): 135.79
Intercept (c): 180616.44
Prediction for 3300 sq ft: $628715.75
```




```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```